Why do I make the distinction between backend and frontend? Well, there are a few examples that fall exactly in one or the other category. For example, Grafana is a pure frontend destination (sometimes also called the *presentation stage*) that queries backend destinations to visualize or alert on signals. The Cloud Native Computing Foundation (CNCF) project Cortex, on the other hand, is a pure backend destination (besides a rudimentary admin Web UI, you only have the API to ingest and query metrics).

Then, there are offerings from dedicated observability vendors, such as Splunk, Datadog, Dynatrace, Honeycomb, New Relic, Lighstep by ServiceNow, AppDynamics by Cisco, and Sumo Logic, which typically combine frontends and backends, providing an integrated experience. In those cases, you can't use the frontend (UI) separately from the backend. There are also mixed cases in the open source space, like CNCF Jaeger (which sports an ingestion pipeline but requires an additional storage system). We will cover these all-in-ones or unified destinations in chapter 6.

Another special case is cloud-provider-native offerings, especially Amazon CloudWatch, Azure Monitor (including Application Insights), and the Google Cloud operations suite, which cover a wide range of features and provide integrations with different backends and frontends.

> **NOTE** Like I pointed out in chapter 1, regarding why I am using the logs, metrics, and traces boundaries to present topics, I am making the frontend–backend distinction mostly on a conceptual level. This should aid in your understanding; however, by now, you know that the reality is much messier and more complicated.

Now, let's get into it. In this chapter we will

- Have a look at some commonalities and general backend terminology
- Review options for and usage of logs backends
- Review options for and usage of metrics backends
- Review options for and usage of traces backends
- Discuss columnar data stores
- Discuss how to go about selecting a backend

Ready? Let's dive in!

5.1 Backend destination terminology

Backends are the source of truth for all your observability questions, be that a human asking them (ad hoc, based on a hunch) or a piece of code (think an alert that queries a backend to evaluate if a certain condition is met). Before we get into the different types of backends and their pros and cons, we will first step back a bit and look at their common characteristics along with some basic terminology we need for the rest of the chapter and book.